

RESEARCH ARTICLE OPEN ACCESS

Compressible Flow Exfoliation of Two-Dimensional Nanomaterials: Insights Into Layer Separation Informed by Gas Dynamics

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Received: 22 August 2025 | **Revised:** 12 January 2026 | **Accepted:** 13 January 2026

Keywords: compressible flow exfoliation (CFE) | high-throughput exfoliation | particle laden flow | supersonic flow | two-dimensional (2D) nanomaterials

ABSTRACT

As the technological implications of 2D nanomaterials like graphene, hexagonal boron nitride (h-BN), and molybdenum disulfide (MoS₂) continue to expand, a critical need arises for the streamlined production of these materials. Their characteristics such as high mechanical robustness, exceptional electrical and thermal conductivities, and substantial surface area effects underscore this urgency. This study delves into an innovative technique, the compressible flow exfoliation (CFE) method, for the aim of achieving scalable production of 2D h-BN nanosheets. The CFE technique capitalizes on the properties of high-pressure gas, laden with 2D nanomaterials, as it traverses an orifice, acting as a converging–divergent (CD) nozzle. In such situations, the supersonic flow results in the formation of shock waves, exerting a sudden drag force on the nanomaterials. However, it is revealed that such shock events alone are not sufficient for separating the 2D layers in nanomaterials. Through detailed particle laden-flow analysis and experiments, we reveal that the key to initiating exfoliation lies in the presence of a thrust force, which is critical for effectively separating these layers. The ensuing flow structure understanding is validated using a CD nozzle and a shock tube experiment, yielding the insight that a thrust force, originating from acceleration at the nozzle throat, is essential for material exfoliation. The rapid and continuous nature of the CFE process signifies its superiority over conventional 2D material exfoliation methods, attributing to environmentally friendly processing, decreased defect emergence, and adaptability to a wide variety of 2D layered materials utilizing a gaseous medium.

1 | Introduction

2D materials have garnered significant attention in recent years due to their unique properties and potential for a wide range of applications. These materials, which include graphene [1], molybdenum disulfide (MoS₂) [2], hexagonal boron nitride (h-BN) [3], exhibit a diverse range of properties that make them particularly advantageous for various applications. These include serving as active materials [4], transistors [5], photodetectors [6], solar cells [7], next-generation electronics [8, 9], energy storage [9–12], water filtration [13], textiles [14], composites [15, 16], and health care [17, 18].

The production of 2D materials involves top-down or bottom-up synthesis methods. Top-down methods like ball milling [19], liquid exfoliation (LPE) [20], micromechanical cleavage [21], and chemical exfoliation [22] are preferred due to their scalability and cost-effectiveness. Micromechanical cleavage utilizes mechanical force to produce 2D flakes from bulk materials, while ball milling exfoliates materials through shear and compressive force [23–25]. LPE involves the use of ionic liquids, surfactants, and organic solvents to exfoliate bulk 2D materials into single or few-layer nanosheets [15] and chemical exfoliation involves oxidation of graphite and thermal expansion of graphene oxide to

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create single layer of graphene. However, these methods have drawbacks such as long processing times, the use of toxic solvents, and the induction of defects due to lattice structural damage. To overcome these challenges, a continuous, high throughput production method called compressible Flow exfoliation (CFE) has recently been developed [26, 27]. This state-of-the-art approach employs high-pressure, high-velocity compressible gases such as helium, nitrogen, or carbon dioxide to carry bulk 2D materials and force them through a narrow orifice. The high shear rates generated by these accelerating gases enable efficient exfoliation, thereby eliminating the need for time-consuming treatments commonly associated with other methods. Notably, CFE utilizes noninert, nontoxic gases, making the process both cost-effective and eco-friendly. The structural integrity and layer-number distribution of CFE exfoliated h-BN, MoS₂, and graphene were thoroughly characterized using Raman spectroscopy, Atomic Force Microscope (AFM), Transmission electron microscopy (TEM), and UV-vis technique. These measurements confirmed that the CFE process yields few-layer nanosheets with minimal defect formation. Furthermore, the technique offers distinct advantages over traditional mechanical methods like ball milling, which relies on high-energy impact forces that often fragment the lattice and induce defects. In contrast, CFE have shown to preserve the in-plane crystallinity of the nanosheets while achieving high production rates. This fundamental difference underlies the superior quality of CFE-produced materials [28, 29].

Despite its advantages, the mechanism underlying the CFE process remains far from being fully understood. Although a high shear rate generated in the confined geometry is known to be responsible for exfoliating 2D nanomaterials, it is unclear why the material yield increases with upstream pressure while the shear rate remains unchanged. This suggests that factors other than shear rate may contribute to the exfoliation process. One possibility is the creation of multiple shock waves within the high supersonic flow generated in the confined geometry, which could influence nanomaterial exfoliation [28, 29]. Further investigation is required to explore the underlying mechanism of the CFE process. Additionally, the high accelerating velocity after the throat or confined section in the CFE geometry may induce forces on the nanomaterials that aid in breaking the in-plane bonding or bonding between layers. However, research on the effect of shock waves on nanomaterials is still in its early stages. Some literature indicates the use of shock waves in sonochemical synthesis through bubble growth, formation, and collapse, or by applying ultrasound [30]. Chen et al. [31] shows shock wave induced formation of CO₂ to few layers graphene through high temperature chemical reaction. A decrease of interlayer spacing and vibration properties of few layers MoSe₂ using high temperature shock wave compression was shown by Vasu et al. [32]. Interestingly, the CFE process shares similarities with cold spray technology, which utilizes high velocity gas flows to achieve specific particle kinetic energies for surface coating. However, in cold spray, supersonic gas flow-induced shock waves may be undesirable as they can cause deceleration and lead to unfavorable coatings [33–35].

Computational methods in materials science are indispensable for investigating the complex behaviors of 2D materials. Techniques such as direct numerical simulations, computational fluid dynamics (CFD) [36–38], molecular dynamics [39, 40], and

density functional theory [41, 42] simulations provide a comprehensive understanding of the properties and separation processes of 2D materials. Specifically, direct numerical simulations focusing on compressible viscous flow around nanomaterials offer valuable insights into the fluid dynamics crucial for the confined fluid exfoliation process. This area of research establishes a fundamental understanding of how fluid dynamics influence nanomaterial separation and exfoliation, opening avenues for enhancing the CFE process.

In this study, we aim to investigate the underlying mechanism of the CFE process using two different setups: a shock tube and a commercial convergent-divergent (CD) nozzle. Both setups create shock waves, but in different ways: the shock tube creates a traveling shock wave inside a tube, while the CD nozzle creates a stationary shock wave inside the nozzle depending on the pressure ratio. Hexagonal boron nitride (h-BN) is used as a representative 2D nanomaterial in both setups. We also employed a quasi-1D model to simplify and solve the complex fluid-solid interaction dynamics affecting the breaking and separation of nanomaterial layers. Additionally, we quantified the mechanical impact on particles by calculating the work done as they traverse through the fluid flow.

2 | Results and Discussion

The exfoliation of 2D materials via CFE is driven by high-velocity gas flow interacting with layered nanomaterials. Understanding the role of gas dynamics, shock waves, and particle forces is crucial for optimizing this process and achieving higher exfoliation yields. Before delving into postcentrifuging specifics, it is essential to establish the underlying mechanisms that govern exfoliation efficiency. The effectiveness of CFE depends on the interaction between the traveling gas flow and layered materials, where forces contribute to the separation of nanosheets. To investigate this, we examine how different experimental setups—such as open and closed shock tube configurations—affect material exfoliation. Additionally, we analyze the flow dynamics within a CD nozzle to assess how variations in gas type and pressure influence particle acceleration and exfoliation performance. UV-vis technique was employed to correlate the gas dynamics study with the exfoliation behavior. The structural characteristics of exfoliated h-BN produced under CFE conditions have been extensively characterized in prior CFE work using Raman spectroscopy, AFM, and TEM [27]. As the present study focuses on elucidating the fundamental flow mechanisms governing the CFE process, repetition of these established structural characterizations was not deemed necessary for the scope of the present work.

2.1 | Materials

In the present study, hexagonal boron nitride (h-BN) was selected as the 2D material of choice due to its ready availability, nontoxic nature, and high degree of purity. The h-BN samples, with a thickness of 13 μm , were procured from Momentive Performance Materials, located in Waterford, NY. As for the solvents employed, isopropanol (IPA) and deionized water were sourced from Fisher Scientific, based in Waltham, MA. Additionally, compressed gases of helium and nitrogen, both

with a purity level of 99.99%, were obtained from Airgas Inc., situated in Radnor, PA. It is noteworthy that all materials and solvents were utilized as received, without any further purification processes.

2.2 | Experimental Setup for Shock Tube and CD Nozzle Studies

The experimental setup for the shock tube in Figure 1 consisted of two 1 in. diameter (1 NPT) polyvinyl chloride (PVC) pipes. The driver section of the tube was 12 in. in length, while the driven section measured 10 in. To facilitate the separation of these two sections, a pair of 1 in. diameter thick-wall PVC flanges with an outer diameter of 4.25 in. were employed. These flanges served to secure varying layers of 0.0045 in.-thick PVC sheets, which acted as the diaphragm.

A pressure gauge was strategically positioned on the upstream side, prior to the driver section, to monitor upstream pressure levels. Given the closed configuration of the shock tube, a pressure release valve was incorporated just before the driver section to enable the release of trapped pressurized gas following diaphragm rupture.

To investigate the influence of standing shock waves, a commercial CD sonic nozzle was incorporated into the setup (Figure 2). This nozzle was procured from FlowMaxx Engineering Inc. and had a length of 2.25 in. (0.057 m) and a diameter of 0.125 in. (0.0032 m). The inlet and outlet diameters of the nozzle were 0.68 in. (0.017 m) and 0.39 in. (0.0098 m), respectively, as depicted in Figure 2a. Nozzle inlet was connected to a material chamber, which was used for feeding the h-BN samples. Subsequently, this chamber was linked to the gas supply line, as illustrated in Figure 2b.

2.2.1 | Experimental Methodology

Prior to the commencement of the shock tube experiments, the driver and driven sections of the shock tube were securely fastened using ½ in. thread size steel hex screws and nuts. A diaphragm, composed of varying layers of 0.0045 in.-thick PVC sheets, was positioned between the two flanges. ~300 mg of h-BN powder was accurately weighed using a Sartorius Cubis precision laboratory balance and introduced into the driven section via a plastic lab spatula.

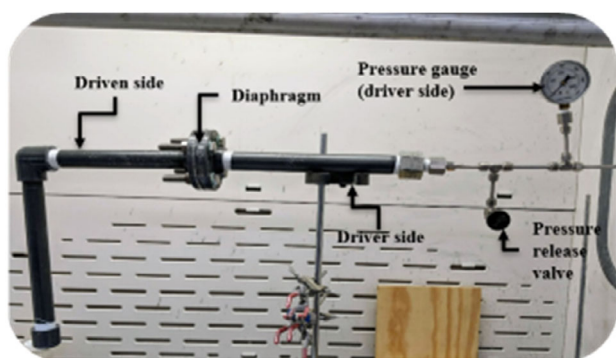


FIGURE 1 | Experimental setup of a shock tube assembly located within a fume hood, highlighting key components.

Initial experiments were conducted using helium gas, followed by comparative trials with nitrogen gas. Both open and closed-end configurations of the shock tube were evaluated. The flow of compressed gas was meticulously regulated via a plug valve. The internal pressure within the driver section varied according to the number and thickness of the diaphragm layers. Observations indicated that 4, 8, and 16 layers of PVC sheets could withstand bursting pressures of 105, 210, and 420 psi, respectively. The abrupt rupture of the diaphragm generated a shock wave, propelling the h-BN powder through the driven section.

For the open-end shock tube experiments, the propelled materials were captured in a collection vessel containing 100 mL of IPA. In the case of CD nozzle experiments, ~300 mg of h-BN powder was loaded into the material chamber. Experiments were conducted at upstream pressures of 200 and 800 psi using both helium and nitrogen gases. The high-pressure gas flow facilitated the transport of materials through the CD nozzle into a boiling flask containing 100 mL of IPA.

In both experimental setups, the gas flow was maintained for a duration of 2 s. The resulting suspensions were collected in two 50 mL conical tubes and subsequently centrifuged at 1400 rpm (rcf = 420 g) for 90 min. This enabled the separation of exfoliated 2D nanomaterials from unexfoliated bulk materials. Postcentrifugation, the heavier bulk materials settled at the bottom, while the lighter exfoliated nanomaterials remained suspended in the solvent. These were carefully extracted using a pipette from the upper layer of the conical tubes.

2.2.2 | Impact of Shock Wave: Experimental Analysis

Experimental investigations of particle-shockwave interactions were carried out using a custom shock-tube setup. The shock-tube setup relies on layered PVC films acting as the diaphragm, and the setup has a removable cap to operate in either open-end or closed-end configurations [43]. As is typical in 2D nanomaterial synthesis studies, the collected product is size separated using high-speed centrifugation [44], during which the “bulk” crystalline material is sedimented while the exfoliated single and few atomic layered nanosheets constitute the colloidal supernatant. The shock tube experiment conducted here resulted in low sample concentrations, adding complexity to the gravimetric analysis needed to measure exfoliated concentrations. To quantify these concentrations, we measure the absorption of exfoliated 2D nanomaterial dispersions (Equation (1)) using a UV-vis spectrophotometer (Shimadzu UV-2450). This device, fitted with a 1 cm path cuvette, scans a wavelength range from 300 to 800 nm at slow intervals of 0.5 nm and employs a slit size of 1.0 Å. We interpret higher light absorbance as interpreted through Beer-Lambert law (Equation (1)) as an indication of superior exfoliation yield and quality.

$$A = \alpha l C \quad (1)$$

where A is the absorbance of the material, α is the molar absorptivity coefficient, l is the optical length of the light through the cuvette, and C is the concentration of the material in mol/L.

The UV-vis absorbance results (Figure 3a) showed that as the shock-tube driver-side pressure was raised to 1.4 MPa, the absorbance values of collected colloids of BN nanosheets in the closed-end shock tube remained almost flat and close to zero.

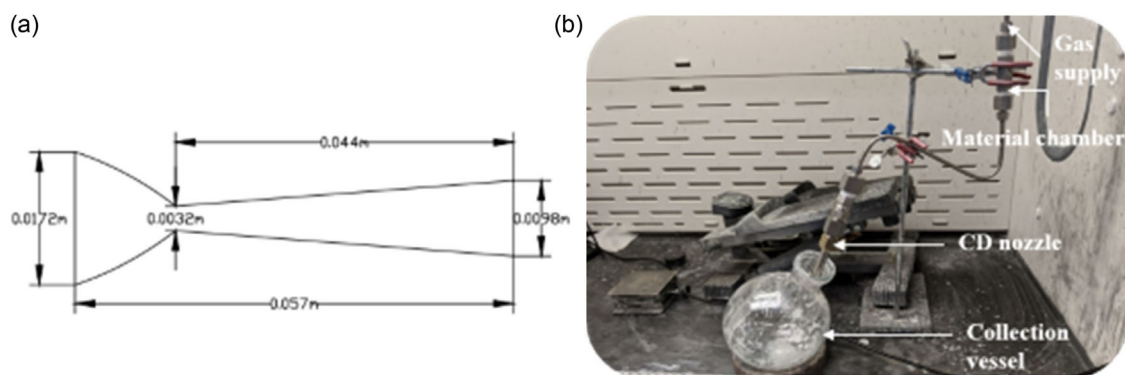


FIGURE 2 | (a) Schematic diagram illustrating the design of a CD nozzle. (b) Experimental configuration of the CD nozzle employed in the CFE process.

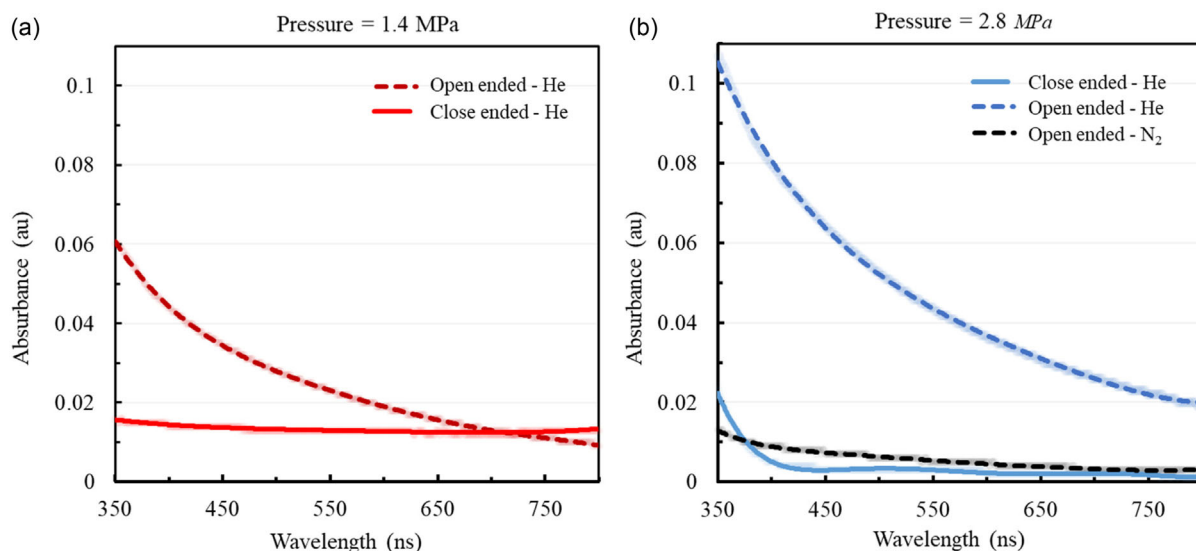


FIGURE 3 | Comparison of UV-vis absorbance of BNNS through open ended and close ended shock tube experiments at different pressure conditions: (a) 1.4 MPa and (b) 2.8 MPa.

In contrast, when the shock tube end was open, the absorbance was nearly four times higher at a reference wavelength of 350 nm. Further, increasing the pressure to 2.8 MPa in the open-ended shock tube resulted in an absorbance that was almost five times higher than its closed-end counterpart. This observation suggests that while the traveling shock wave inside the tube aided in the exfoliation of BN nanosheets, an exit thrust force may be necessary to achieve higher yields. Furthermore, increase to the driver-side pressure aided exfoliation, where the absorbance of the nanosheets at 2.8 MPa was approximately two times higher than at 1.4 MPa (Figure 3a,b).

To understand the shock tube parameters and their influence on the exfoliation process of 2D materials, various shock tube parameters were predicted based on the relationships established in the literature [45, 46]. Table 1 indicates that increasing the pressure from 0.8 to 2.8 MPa will result in an increase in the shock Mach number from ~ 1.9 to ~ 2.7 , and the shock velocity from ~ 636 to ~ 921 m/s. Notably, a maximum Mach number of 4 could be achieved by further increasing the pressure to 2.8 MPa, suggesting the potential for higher yields of nanosheets. However, due to the rated pressure limitation of the PVC tube,

the experimental investigation was constrained to a maximum pressure of 2.8 MPa. Additional details on shock tube relations using helium at different upstream pressures can be found in Table 1.

To compare the UV-vis absorbance between different gases, nitrogen gas was employed alongside helium gas in the experimental shock-tube setup. As anticipated, the results indicated that helium resulted in absorbance levels ~ 11 times higher at a wavelength of 350 nm compared to nitrogen gas in the open-end shock tube at 2.8 MPa. This stark difference is attributed to the distinct gas dynamics of the two species. As quantitatively demonstrated in our computational analysis (Figure 4), Helium is a lighter monoatomic gas that achieves significantly higher flow velocities ($\sim$ Mach 10) compared to Nitrogen ($\sim$ Mach 2). This substantial difference in kinetic energy, confirmed by the simulation data, directly correlates with the increase in absorbance observed here.

The UV-vis absorbance experimental findings clearly demonstrated that the closed-end shock tube configuration yielded poor material yield. This suggests that the traveling shock wave passing through the material sample does not significantly contribute

TABLE 1 | Postshock wave characteristics in a helium gas-filled shock tube: analyzing upstream pressure (P_u), driver pressure (P_d), shock Mach number (M_s), shock velocity (V_s), temperature behind the shock wave (T_b), reflected pressure (P_r), Mach number behind the incident shock (M_b), and reflected Mach number (M_r).

P_d , MPa	P_u , MPa	M_s	V_s , ms^{-1}	T_b , K	P_r , MPa	M_b	M_r
0.8	0.1	1.9	636	451.1	1.2	0.89	1.78
1.4	0.1	2.3	765	542.0	2.2	1.10	0.81
2.8	0.1	2.7	921	670.6	3.8	1.28	1.32
5.6	0.1	3.2	1077	822.9	5.9	1.41	2.03
11	0.1	3.7	1255	1023.6	8.9	1.51	2.63
13.8	0.1	3.9	1314	1096.0	10	1.54	3.51
16.5	0.1	4.0	1362	1158.3	10.9	1.56	2.93

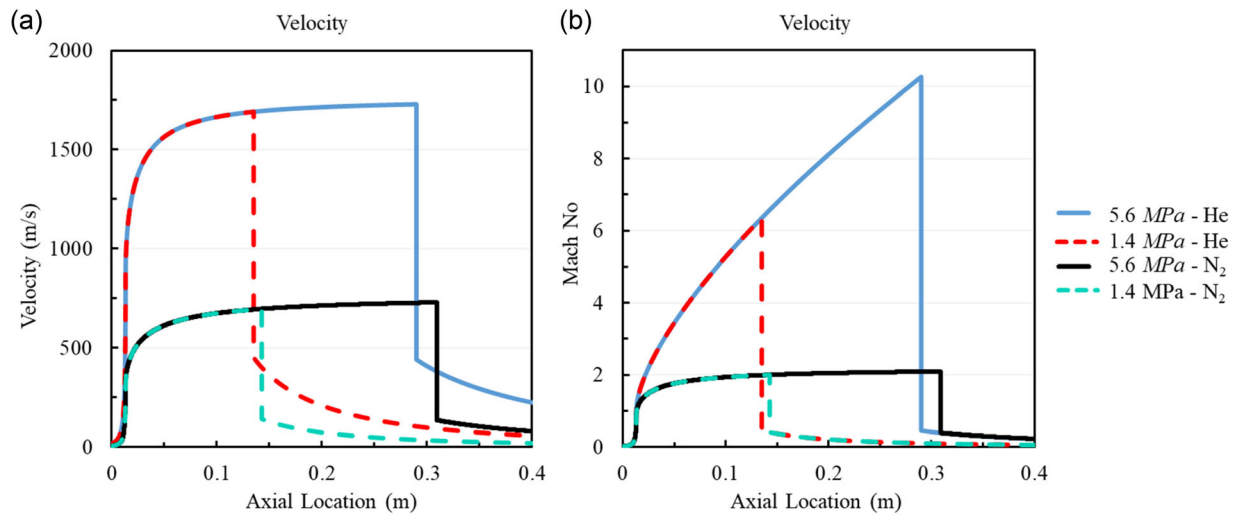


FIGURE 4 | (a) Velocity profile and (b) Mach number profile through the CD nozzle using Quasi 1D Euler equations.

to the exfoliation process. Conversely, the open shock tube configuration exhibited significantly higher absorbance of the materials, indicating the crucial role of particles and accelerating fluid flow at the exit of the shock tube. Furthermore, the use of higher pressures achieved by employing helium, a faster monoatomic gas, resulted in enhanced absorbance and yields of nanomaterials compared to nitrogen, a slower diatomic gas, in the open-end shock tube configuration.

These results emphasize the importance of investigating the impact of standing shock waves as a result of flow in a CD nozzle that may be encountered during the CFE process on 2D materials. Additionally, the necessity of high-speed fluid and particle flow in the exfoliation process warrants further exploration. Therefore, prior to conducting experimental investigations in CD nozzle flow, comprehensive computational work was first performed to analyze the fluid dynamics and flow structure.

2.2.3 | Flow Dynamics and Shock Wave Analysis: Quasi 1D Analysis

The Quasi 1D Euler equations were utilized to accurately capture the steady shock wave phenomena occurring within the nozzle [47]. The use of the Quasi 1D Euler code served as a valuable tool to confirm the accuracy and reliability of the simulation outcomes. Specifically, it facilitated the capture of the shock wave

behavior and provided a comprehensive understanding of the particle-laden flow characteristics within the nozzle during the exfoliation process.

The Quasi 1D Euler analysis was based on considering the following 1D Euler equations

$$\frac{\partial}{\partial t} \begin{bmatrix} \rho \\ \rho u \\ E_t \end{bmatrix} + \frac{\partial}{\partial x} \begin{bmatrix} \rho u \\ \rho u^2 + p \\ \rho u H \end{bmatrix} = - \frac{1}{A} \frac{\partial A}{\partial x} \begin{bmatrix} \rho u \\ \rho u^2 \\ \rho u H \end{bmatrix} \quad (2)$$

where H is total enthalpy, $H = (E_t + p)/\rho$, E_t is the total energy, $E_t = \frac{p}{\gamma-1} + \frac{1}{2}u^2$, p is the total pressure, ρ is the density of fluid, γ is the ratio of specific heats at constant pressure (C_p) and constant volume (C_v), $\gamma = \frac{C_p}{C_v}$. The velocity and Mach number plots from the resulting solutions were generated in Figure 4 for a comparative analysis of the four cases: helium at 5.6 and 1.4 MPa, and nitrogen at 5.6 and 1.4 MPa, using Equation (2). To validate the Quasi 1D predictions, simulation was performed using COMSOL Multiphysics 5.3a for upstream pressure of 5.6 MPa described in Supporting Information. The quasi-1D Euler approach was used for subsequent analysis because COMSOL could not adequately resolve the shock and boundary-layer features within the nozzle due to mesh-resolution limitations, whereas the quasi-1D model reliably captured the essential flow trends needed for this study (Figure S3).

As shown in Figure 4a, for helium at 5.6 MPa, a maximum velocity of ~ 1730 m/s was achieved, corresponding to a Mach number of ~ 10 (Figure 4b). This high velocity was sustained up to a distance of ~ 0.3 m along the CD nozzle before encountering a normal shock wave. In the case of helium at 1.4 MPa, a maximum velocity of ~ 1690 m/s was attained, resulting in a Mach number of about 6. However, due to the lower upstream stagnation pressure, the shock wave occurred at a distance of ~ 0.15 m inside the nozzle.

In contrast, nitrogen at both 5.6 and 1.4 MPa exhibited lower velocities. Nitrogen at 5.6 MPa achieved a maximum velocity of ~ 730 m/s, while nitrogen at 1.4 MPa reached a maximum velocity of ~ 700 m/s. The corresponding Mach numbers for both cases were ~ 2 . Notably, nitrogen at 5.6 MPa traveled a further distance of up to ~ 0.3 m before encountering a shock wave, compared to nitrogen at 1.4 MPa, which experienced the shock wave at a distance of ~ 0.15 m.

These results demonstrate the significant influence of gas properties and pressure on the velocity and Mach number profiles within the CD nozzle. Helium, being a lighter monoatomic gas, exhibited higher velocities and Mach numbers compared to nitrogen, which is a heavier diatomic gas. Additionally, variations in pressure affected the flow characteristics and the location at which the shock wave formed within the nozzle.

The analysis presented in Figure 4 provides crucial insights into the flow dynamics and shock wave behavior for different gas configurations and pressures. These findings contribute to a better understanding of the exfoliation process and enable the optimization of CD nozzle-based experimental conditions to achieve greater yields.

2.2.4 | Exfoliation Efficacy Under Varying Gas Conditions

The BN nanosheets obtained from the CD nozzle experiments were handled and characterized in a similar manner to the shock-tube experiments. Furthermore, UV-vis absorbance measurements were also conducted in a similar protocol to evaluate the yield performance of different gas configurations in the CFE

process using CD nozzle. Figure 5b presents the mean absorbance values from three independent exfoliation trials for each gas-pressure condition, with error bars representing ± 1 standard deviation.

Among the various cases investigated, flowing helium at 5.6 MPa resulted in the highest absorbance as well as greater variability, indicating better exfoliation of the nanosheets (Figure 5a), compared to the other experimental conditions studied. Moreover, the absorbance at a specific wavelength (300 nm) [48, 49] was compared between helium sprayed at 1.4 MPa and nitrogen sprayed at a significantly higher pressure of 5.6 MPa. Despite the huge pressure difference, the absorbance was found to be higher when helium at 1.4 MPa was utilized, indicating a more efficient exfoliation process (Figure 5b) with the lighter carrier gas. The accompanying narrow standard deviation further supports the consistent performance of helium at 1.4 MPa, as indicated by its higher absorbance value and lower variability compared to nitrogen at 5.6 MPa.

These results highlight the importance of gas selection and pressure in achieving optimal exfoliation of h-BN nanosheets. The use of helium at 5.6 MPa demonstrated superior performance, resulting in higher absorbance and improved consistency compared to other gas configurations tested.

The difference in absorbance can also be observed by utilizing the Tyndall effect, where laser light is passed through the vial containing the sample suspended in a solvent [50]. As shown in Figure 6, it is evident that the sample prepared at helium pressure of 5.6 MPa exhibits higher visibility, while the visibility becomes fainter in the case of helium at 1.4 MPa and further diminished in the cases of nitrogen at 5.6 and 1.4 MPa.

The dispersion stability of exfoliated h-BN at 800 psi was evaluated using time-dependent UV-vis spectroscopy. Figure 7 shows the absorbance spectra recorded immediately after centrifuging the exfoliated solution (0 h) and after 12, 24, and 48 h of rest. All spectra exhibit a featureless, monotonically decreasing absorbance profile characteristic of liquid-phase dispersed 2D nanosheets. While a gradual reduction in absorbance intensity is observed over time, the overall spectral shape remains

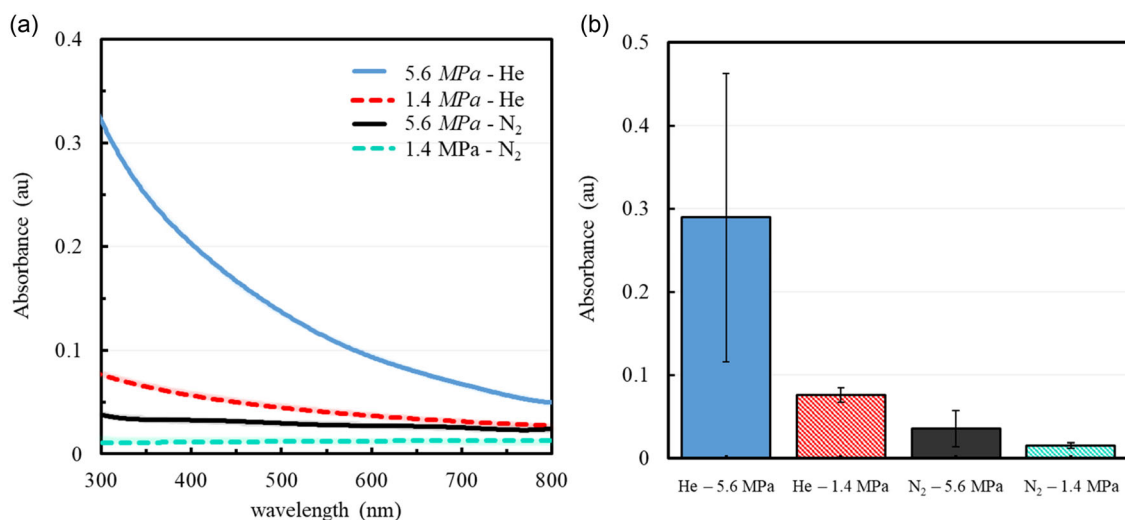


FIGURE 5 | (a) UV-vis absorption of h-BN sprayed through helium and N₂ at different pressures through CD nozzle, and (b) bar graph comparing the UV-vis absorbance at 300 nm for different pressures.

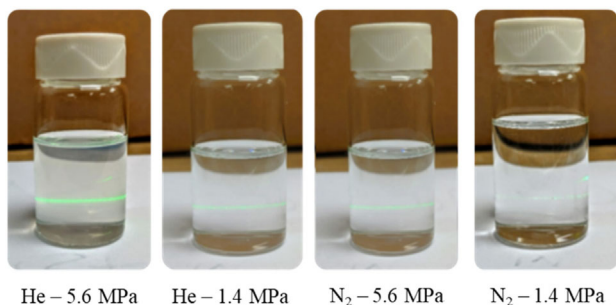


FIGURE 6 | Exfoliated suspensions of BN nanosheets at different upstream pressures using helium and nitrogen.

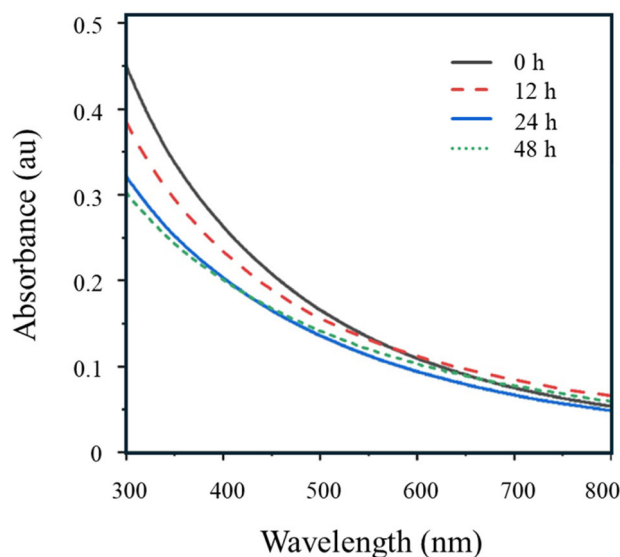


FIGURE 7 | Time-dependent UV-vis absorbance spectra of exfoliated h-BN dispersions prepared by CFE process at helium pressure of 800 psi through CD nozzle. Spectra recorded at 0–48 h after centrifugation illustrates the dispersion stability over time.

unchanged, indicating the absence of chemical degradation or aggregation-induced spectral features. Notably, significant absorbance persists even after 48 h, demonstrating good dispersion stability with only limited sedimentation of larger flakes. It has been reported that there is a stable suspension even after 6 months of storage by CFE process.

The few-layer nature of h-BN obtained after CFE processing and centrifugal separation was confirmed by Raman spectroscopy of drop-cast flakes. Figure S1 shows Raman spectra collected using a 633 nm excitation. Bulk h-BN exhibits a characteristic E_{2g} phonon mode at $\sim 1366 \text{ cm}^{-1}$ whereas CFE-processed h-BN exhibits a pronounced redshift of the E_{2g} peak to $\sim 1362 \text{ cm}^{-1}$ indicating a significant presence of few-layer and bilayer h-BN.

The structure of BN nanoparticles after CFE processing was characterized by AFM, with a representative flake shown in Figure S2. The flake exhibits a thickness of $\sim 2 \text{ nm}$ and a lateral width of $\sim 350 \text{ nm}$, corresponding to an aspect ratio of ~ 175 . AFM analysis of CFE process has shown that higher driving pressures generally lead to improved flake quality. BN exfoliated using helium at 5.6 MPa exhibits an average thickness of 5.7 nm and a lateral size of 160 nm, corresponding to an

aspect ratio of 28. In addition, the fraction of few-layer flakes (≤ 10 layers) increases from $\sim 27\%$ at 5.5 MPa to 43% at 10 MPa, demonstrating enhanced exfoliation efficiency at higher pressures. However, the lower exfoliation yield observed here compared to the original CFE process is primarily due to differences in nozzle geometry. The CD nozzle used in this study has a larger throat diameter (3.2 mm) than the $\sim 0.1 \text{ mm}$ opening employed in CFE. This larger throat may reduce gas acceleration, shear, and gas dynamics at a given pressure, thereby lowering exfoliation efficiency.

Nonetheless, this analysis suggests that there are specific forces or energy requirements that must be overcome for breaking the interlayer bonding and in-plane bonding within the 2D materials as they traverse through the CD nozzle at higher dynamic pressures. These requirements can be further analyzed in the context of the forces and energy transfer experienced by the particles occurring during supersonic particle-laden flows.

2.2.5 | Force Dynamics Analysis

This section primarily examines the forces responsible for breaking interlayer and in-plane bonds within 2D materials as they traverse the CD nozzle under high-velocity gas flow conditions. It's crucial to understand that energy thresholds must be surpassed to facilitate this disruption process. The influence of these forces on the nanomaterials can be comprehensively evaluated using the drag equation [35]

$$F_D = \frac{1}{2} C_D \rho A (dV)^2 \quad (3)$$

where F_D represents the drag force, ρ is the fluid density, A is the effective cross-sectional area of the particle, C_D is the drag coefficient, and dV is the relative velocity between the fluid and the particle.

The drag force calculation provides insights into the forces acting on the nanomaterials during their passage through the CD nozzle. These forces play a crucial role in the exfoliation process by breaking the interlayer and in-plane bonding. Since the starting bulk particles, prior to exfoliation, are assumed to be spherical, the effective cross-sectional area (A) can be calculated using the formula $A = \frac{\pi d_p^2}{4}$, where d_p represents the diameter of the particle. For the h-BN sample particles, the diameter is assumed to be $13 \times 10^{-6} \text{ m}$, as provided by the h-BN supplier. The particle density (ρ) is assumed to be 2100 kg/m^3 , representing the density of h-BN particles.

In Equation (3), C_D represents the drag coefficient [36], which varies depending on the particle Reynolds number R_N . The particle Reynolds number is calculated using the formula $R_N = \rho (dV) \frac{d_p}{\mu}$, where μ represents the dynamic viscosity of the fluid.

The particle Reynolds numbers obtained during the flow at different upstream pressures (1.4 MPa, 5.6 MPa) using both helium and nitrogen gases are presented in Figure S4. The flow properties, such as density and viscosity, have been considered to determine these particle Reynolds numbers. It is worth mentioning that even higher velocities achieved by helium, its Reynolds number is lower compared to nitrogen due to higher kinematic viscosity of helium.

The fluid velocities for the four different cases, namely helium at 1.4 MPa, helium at 5.6 MPa, nitrogen at 1.4 MPa, and nitrogen at 5.6 MPa, have been determined using the Quasi 1D Euler equation. This computational approach allows for the estimation of fluid flow behavior and dynamics within the CD nozzle.

To calculate the particle velocities, the time marching method has been employed, utilizing the 2N-storage low dissipation and dispersion Runge–Kutta scheme [37]. This numerical scheme enables accurate and efficient computation of particle velocities based on the governing equations and boundary conditions.

By combining the fluid velocities obtained from the Quasi 1D Euler equation with the computed particle velocities using the time marching method, a comprehensive understanding of the flow characteristics and the behavior of particles within the CD nozzle can be achieved. These calculations contribute to the overall analysis of the exfoliation process and help elucidate the influence of fluid dynamics on the exfoliation efficiency.

Figure 8 shows the results of the Quasi 1D Euler analysis of a particle-laden flow in a CD nozzle. Figure 8a shows the relative velocity, which is defined as the difference between the flow velocity and the particle velocity. Similarly, Figure 8b shows the gas density changes during the same flow profile. According to Figure 8a, just after the throat of the CD nozzle, situated at a distance of 0.013 m from datum, is where the largest difference between the flow velocity and particle velocity (peak relative velocity) is achieved. After this peak, the relative velocity stabilizes until the flow and particles encounter the normal shock wave (around 0.14 and 0.3 m from the nozzle inlet) resulting in a sudden decrease in the relative velocity in the opposite direction. It is worth mentioning that the Quasi 1D Euler equation, being a 1D model, predicts a normal shock wave rather than an oblique shock wave, which is of more frequent occurrence in supersonic flows. The fluid density as measured during the same cross-sectional flow profile (shown in Figure 8b) undergoes a decrease by several orders of magnitude as the flow accelerates followed by sudden increase in density at the point of normal shock. For all four cases of flow configurations considered,

the density change experienced postshock is only about one order of magnitude higher. Contrast this with preshock flow, where the density of the fluid decreases by up to three orders of magnitude.

The drag force that a particle experience is shown in Figure 9 and it depends on the relative velocity between the fluid and particles, as well as local fluid density around the particle, as indicated in Equation (3). In the case of helium gas at an upstream pressure of 5.6 MPa, the particles experience a peak drag force of 225 μN just past the throat, and well before the occurrence of the shock wave. However, when these particles confront a shock wave, the drag force drops to 50 μN but in the opposite direction. When the upstream pressure is reduced to 1.4 MPa, the drag force after the throat is 90 μN and within the shock wave region, it changes to a lower value of 30 μN in the opposite direction. For nitrogen gas at an upstream pressure of 5.6 MPa, the particles face a maximum drag force of 100 μN just beyond the throat. This force changes to 50 μN in the opposite direction when encountering a shock wave. At a lower upstream pressure of 1.4 MPa, the drag force after the throat is 40 μN and remains almost the same within the shock wave region. Consequently, it can be observed that helium gas at 5.6 MPa exerts the highest force on the particle among the four cases. This finding indicates that higher pressures and lighter gases can impose greater forces on the particles due to the higher velocity and density changes during the flow. The work done on a particle as it traverses the flow, can be approximated by multiplying the forces on the particle throughout the length of the nozzle by the particle velocity and then integrating over time using the following equation

$$W_p = \int F_D \cdot V_p dt \quad (4)$$

where F_D is the drag force and V_p is the particle velocity.

The cumulative work done on particles for the four different cases (helium at upstream pressures of 1.4 MPa, 5.6 MPa, and nitrogen at 1.4 MPa, 5.6 MPa) is presented in Figure 10, allowing

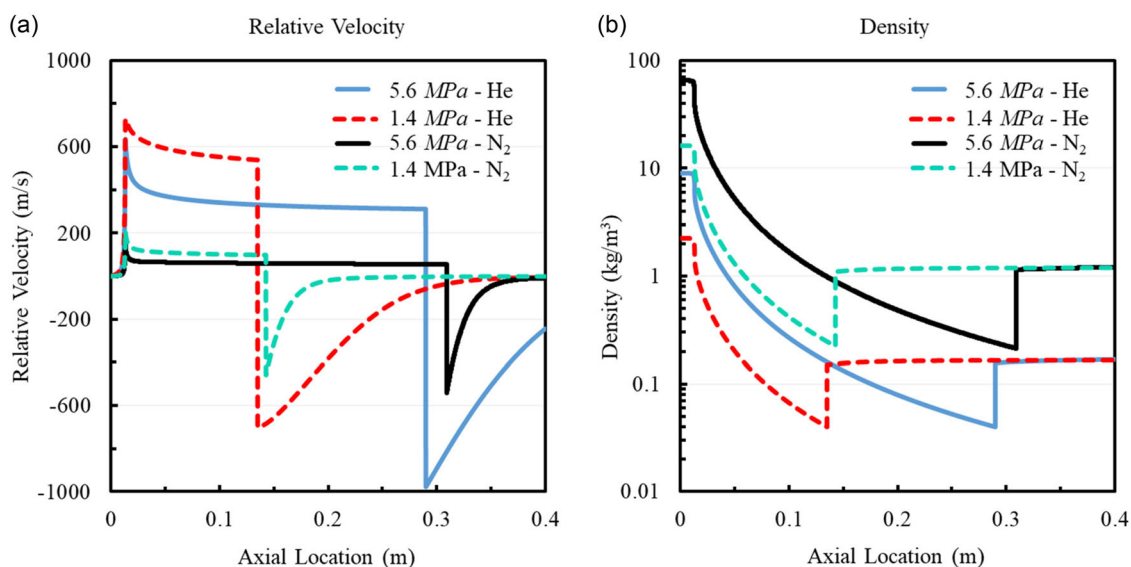


FIGURE 8 | (a) Relative velocity and (b) density variations through the axial distance of the CD nozzle for different upstream pressures of He: 5.6 and 1.4 MPa, and N₂: 5.6 and 1.4 MPa.

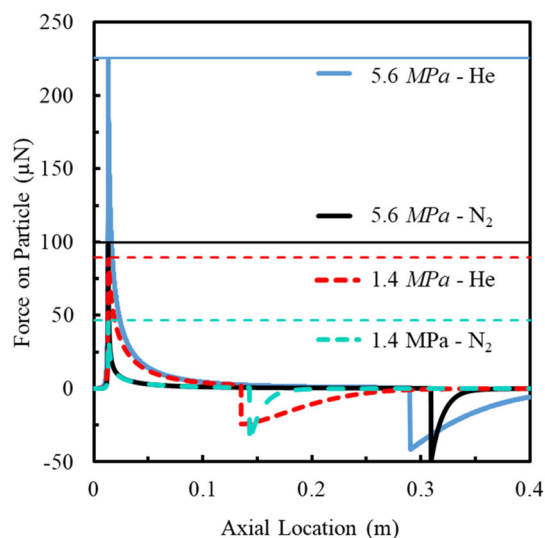


FIGURE 9 | Drag forces calculated on a particle passing through a CD nozzle at different upstream pressures of helium and nitrogen gas. The drag forces represent the resistance experienced by the particle due to the relative velocity between the fluid and the particle. The horizontal lines indicate the maximum drag force exerted on the particles within the CD nozzle at each pressure, which determines the peak force available to initiate bond breakage.

for a comparison. Among the cases, the maximum work done on a particle is $\sim 2.4 \mu\text{J}$ for helium at 5.6 MPa. This value is higher than the maximum work done in the other three cases due to the higher velocity and density changes experienced at the throat and after the shock wave. The maximum work done by helium at 1.4 MPa, nitrogen at 5.6 MPa, and nitrogen at 1.4 MPa are ~ 1.6 , 0.6 , and $0.4 \mu\text{J}$, respectively.

Moreover, it is noteworthy that the graph in Figure 10a demonstrates the effects of faster flow characteristics of helium compared to nitrogen. Helium at 5.6 MPa takes only 0.45 ms to

reach the nozzle exit, whereas nitrogen at 1.4 MPa exhibits the slowest movement, taking $\sim 3.6 \text{ ms}$ to reach the exit.

A common approach for understanding layer separation of 2D materials has been to start from first principles and generate a model particle and compute the energetics of bond breakage [51]. However, agreements with experimental results are often weak [52] due to the polydisperse geometries and characteristics of layered particles and their exfoliated nanomaterials, along with inability to capture the complex dynamics of energy transfer during exfoliation. Instead, a simpler approach could be to rely on the concept of cut-off forces, which is rooted in the requirement that a minimum force (stress) must be exerted for breaking bonds (yielding). Any force applied to a particle greater than the cut-off force will contribute to bond breakage and layer separation (exfoliation). While a force below this threshold will only result in the particle's elastic deformation and other dissipation mechanisms that exclude bond breakage and layer separation. Although unknown, this required minimum force for layer separation can be gleaned from the experimental results and flow analysis presented thus far. Figure 10b depicts the cumulative work (as calculated from Equation (4)) exerted on particles, but only after various cut-off force thresholds (x-axis) – within the range of 0 – $100 \mu\text{N}$ – have been overcome. Within the examined range of Figure 10b, helium at 5.6 MPa shows significantly higher work done on the particle compared to the other three scenarios. This observation is consistent with the experimental results of UV–vis absorbance data previously presented in Figure 5. The maximum cut-off force of $100 \mu\text{N}$ investigated here, results in zero-work being done on the particles in the remaining three scenarios, but not for helium at 5.6 MPa, explaining why exfoliation is so efficient in this condition.

To determine the minimum cut-off force required for the exfoliation of nanomaterials, the ratio of the work done on a particle for each case is compared with the ratio of the UV–vis absorbance results at 5.6 and 1.4 MPa, considering both helium

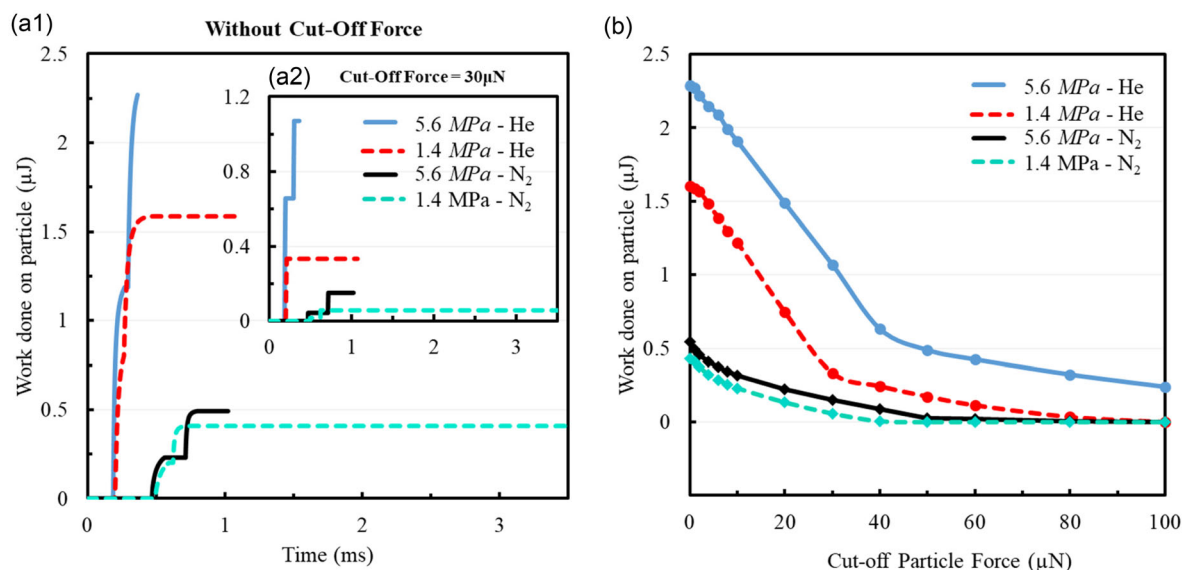


FIGURE 10 | (a) Cumulative work done on a nanoparticle at different upstream pressures using helium and nitrogen gas (a1) without cut-off force and (a2) at cut-off force = $30 \mu\text{N}$. The cumulative work done represents the total energy exerted on the particle throughout its trajectory through the CD nozzle. (b) Work done on the particle at different cut-off forces. The graph illustrates the work done on the particle at various threshold forces, indicating the minimum force required to initiate the exfoliation of nanomaterials during compressible flows.

and nitrogen gases. The absorbance results at a wavelength of 300 nm are used for this calculation, as it is commonly employed as a reference wavelength to estimate concentrations and process yields [48, 49]. Further analysis of these ratios can provide insights into the relationship between the work done on particles and the resulting exfoliation efficiency, enabling the determination of optimal conditions for effective nanomaterial exfoliation.

The ratio of UV–vis absorbance results and the work ratio at different cut-off forces are presented in Tables S1 and S2. Notably, the simulated work ratio value at 30 μN of Table S2 closely matches the actual absorbance ratio values in Table S1. This alignment suggests that a minimum cut-off force of $\sim 30\ \mu\text{N}$ is sufficient to initiate the exfoliation of nanomaterials during compressible flows. This finding is consistent with experimental measurements [53] and theoretical calculations [54], which indicate that weak Van der Waals interactions between the layers do not require a large force for exfoliation. It is reported that slip between layers in a graphite flake with a lateral size of 10 μm occurs at a small shear force of 3.9 μN [53]. However, investigations into interlayer sliding in h-BN reveal that the shear strength of an h-BN bilayer is approximately five times greater than that of a graphene bilayer [55]. The plot in Figure 10a2 at this cut-off force of 30 μN provides a visualization of the work done on the particles under conditions that are favorable for initiating the exfoliation process. By examining the stepwise nature of the plots of cumulative work done at this specific cut-off force in Figure 10a2, it can be seen that certain flow configurations (i.e., helium at 5.6 MPa and nitrogen at 5.6 and 1.4 MPa) impart sufficient energetics for layer separation during the flow acceleration (preshock) and flow deceleration (postshock) stages. While other configurations (i.e., helium at 1.4 MPa) can only impart sufficient energetics for layer separation during the flow acceleration (preshock) stage. This indicates that the preshock flow where the densities due to higher upstream (stagnant) pressures can contribute further to particle drag (thrust) forces and hence to the work done on particles rather than postshock flows. This indicates that optimized nozzle design meant to maximize particles forces can result in effective nanomaterial exfoliation during compressible flows.

Based on the force dynamics and energy thresholds established above, we propose that the dominant exfoliation mechanism in CFE is acceleration-driven aerodynamic shear. As the carrier gas accelerates to high Mach numbers (Mach ~ 10 for Helium) at the nozzle throat, the relatively heavier h-BN particles lag behind due to inertia, creating a significant relative velocity (dv) between the fluid and the solid. This relative motion generates substantial viscous drag forces on the particle surface. Unlike the instantaneous pressure spike of a shock wave, which acts compressively, this drag force acts tangentially along the particle surface. When this aerodynamic shear force exceeds the cut-off threshold of $\sim 30\ \mu\text{N}$, it overcomes the Van der Waals forces holding the layers together, causing the outer nanosheets to slide off the bulk crystal. This model explains why Helium, with its higher velocity and resulting thrust capability, significantly outperforms Nitrogen, even though Nitrogen might generate different shock structures.

3 | Conclusion

In conclusion, this study investigated the mechanism of the CD nozzle CFE process using gas dynamics principles and CFD

analysis. By comparing different cases, it was found that a traveling shockwave alone is insufficient to cause the separation of 2D layers in nanomaterials. Instead, it was determined that the presence of a thrust force is crucial for initiating the exfoliation of nanomaterials. The analysis revealed that both negative drag forces and positive thrust forces are exerted on nanoparticles during the flow process through the CD nozzle. The use of monoatomic, lighter, and faster gases such as helium resulted in higher forces at higher pressures compared to diatomic gases like nitrogen. This higher energy imparted on the nanomaterials led to a greater yield of single or few layers of nanomaterials being separated from the bulk material, as confirmed by UV–vis absorption measurements. It was determined that a minimum force of $\sim 30\ \mu\text{N}$ is required to break the bulk multilayers and initiate the exfoliation process.

The study of isentropic expansion through the CD nozzle, shock waves, drag forces, and energy on nanomaterials provides valuable insights for optimizing the production of nanomaterials. By understanding the relationship between gas properties, flow dynamics, and the forces exerted on the nanoparticles, it is possible to tailor the CFE process for desired outcomes. Finally, considering the economic implications, we define a relative optimal range based on application needs. For high-value industries requiring maximum yield, Helium (at pressures, 1.4 MPa) is the requisite technical optimum. However, for cost-sensitive applications, Nitrogen at 5.6 MPa serves as the economic optimum; while its yield is significantly lower, it provides sufficient thrust to marginally exceed the exfoliation threshold, offering a viable, low-cost production route without the need for expensive helium recovery systems. Overall, this research enhances our understanding of the CFE process and provides insights into the forces and conditions necessary for efficient nanomaterial exfoliation. These findings contribute to the development of improved manufacturing processes for producing high-quality nanomaterials with controlled layer separation.

Author Contributions

Md Farhadul Islam: investigation (lead), methodology (equal), validation (equal), visualization (equal), writing – original draft (equal). **Saeed Arabha:** investigation (equal), methodology (equal), validation (equal), visualization (equal), writing – original draft (equal). **Ray Hixon:** investigation (equal), project administration (equal), supervision (lead), writing – review & editing (equal). **Reza Rizvi:** project administration (lead), supervision (lead), writing – review & editing (equal).

Acknowledgments

This work was supported by the US National Science Foundation's Nanomanufacturing Program (CMMI Award # 1762507), as well as the Canadian National Science and Engineering Research Council (NSERC) Discovery Grant Program (Award # RGPIN-2019-06345).

Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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Supporting Information

Additional supporting information can be found online in the Supporting Information section. **Supporting Fig. S1:** Raman spectra for BN produced using compressible Flow exfoliation (CFE). **Supporting Fig. S2:** The AFM scan image of a BN flake isolated after CFE showing the height profile of the flake and the thickness distribution of several flakes. **Supporting Fig. S3:** (a) Comparison of axial flow velocity profiles at 5.6 MPa obtained from COMSOL Multiphysics and the quasi-1D Euler model using the same CD-nozzle geometry. (b) Corresponding Mach number distributions for both methods. **Supporting Fig. S4:** (a) Particle Reynold's number obtained using helium and (b) nitrogen at different upstream pressures. **Supporting Table S1:** UV-vis absorbance ratio at 300 nm wavelength. **Supporting Table S2:** Work ratio at different cut-off forces.